```python
import numpy as np
import pandas as pd
```

✓ 0.0s Python

1 Create a dataframe to display the result [Attached as screenshot] Problem

```python
data = {
    "Student": ["David", "Samuel", "Terry", "Evan"],
    "Age": [27, 24, 22, 32],
    "Country": ["UK", "Canada", "China", "USA"],
    "Course": ["Python", "Data Structures", "Machine Learning",
               "Web Development"],
    "Marks": [85, 72, 89, 76],
}
df = pd.DataFrame(data)
print("Problem 1:")
print(df, end="\n\n")
```

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```
Problem 1:
  Student  Age Country           Course  Marks
0   David   27      UK           Python     85
1  Samuel   24  Canada   Data Structures     72
2   Terry   22   China  Machine Learning     89
3    Evan   32     USA   Web Development     76
```

2: Retrieve the Marks column and assign it to a variable b

```
b = df["Marks"]
print("Problem 2:")
print(b, end="\n\n")
```

[14] ✓ 0.0s

```
Problem 2:
0    85
1    72
2    89
3    76
Name: Marks, dtype: int64
```

Problem 3: Retrieve the Country and Course columns and assign it to a variable c

```
c = df[["Country", "Course"]]
print("Problem 3:")
print(c, end="\n\n")
```

[15] ✓ 0.0s

```
Problem 3:
  Country           Course
0      UK           Python
1  Canada   Data Structures
2   China  Machine Learning
3     USA   Web Development
```

Problem 4: Data Loading & Exploration: Create a dataset containing at least 5 columns (e.g., Customer_ID, Name, Age, Gender, Purchase_Amount) and 10 records. Load the dataset into a Pandas DataFrame. Display the first 5 rows using head(). Get summary statistics using describe(). Check for missing values and handle them appropriately.

```python
customers = pd.DataFrame({
    "Customer_ID": [101, 102, 103, 104, 105, 106, 107, 108, 109, 110],
    "Name": ["Alice", "Bob", "Charlie", "Diana", "Ethan",
             "Fiona", "George", "Hannah", "Ian", "Julia"],
    "Age": [28, 35, 42, np.nan, 31, 25, 47, 38, 29, 33],
    "Gender": ["Female", "Male", "Male", "Female", "Male",
               "Female", "Male", "Female", None, "Female"],
    "Purchase_Amount": [250.0, 480.5, 320.0, 150.75, np.nan,
                        410.0, 275.25, 530.0, 199.99, 365.5],
})

print("Problem 4:")
print("First 5 rows:")
print(customers.head(), end="\n\n")

print("Summary statistics:")
print(customers.describe(), end="\n\n")

print("Missing values per column:")
print(customers.isnull().sum(), end="\n\n")

# Handle missing values (assign back instead of using inplace=True)
customers["Age"] = customers["Age"].fillna(customers["Age"].median())
customers["Purchase_Amount"] = customers["Purchase_Amount"].fillna(
    customers["Purchase_Amount"].mean()
)
customers["Gender"] = customers["Gender"].fillna(
    customers["Gender"].mode()[0]
)

print("Missing values after handling:")
print(customers.isnull().sum(), end="\n\n")
```

```
Problem 4:
First 5 rows:
   Customer_ID     Name   Age  Gender  Purchase_Amount
0          101    Alice  28.0  Female           250.00
1          102      Bob  35.0    Male           480.50
2          103  Charlie  42.0    Male           320.00
3          104    Diana   NaN  Female           150.75
4          105    Ethan  31.0    Male              NaN

Summary statistics:
       Customer_ID        Age  Purchase_Amount
count     10.00000   9.000000         9.000000
mean     105.50000  34.222222       331.332222
std        3.02765   7.084804       126.898213
min      101.00000  25.000000       150.750000
25%      103.25000  29.000000       250.000000
50%      105.50000  33.000000       320.000000
75%      107.75000  38.000000       410.000000
max      110.00000  47.000000       530.000000

Missing values per column:
Customer_ID        0
Name               0
Age                1
Gender             1
Purchase_Amount    1
dtype: int64

Missing values after handling:
Customer_ID        0
Name               0
Age                0
Gender             0
Purchase_Amount    0
dtype: int64
```

Problem 5: Data Filtering & Selection: Retrieve all customers whose Age is greater than 30. Find the total Purchase Amount for male and female customers separately. Select only the Customer_ID, Name, and Purchase_Amount columns.

```
age_above_30 = customers[customers["Age"] > 30]
print("Problem 5: Customers with Age > 30")
print(age_above_30, end="\n\n")

#
total_by_gender = customers.groupby("Gender")["Purchase_Amount"].sum()
print("Problem 5: Total Purchase Amount by Gender")
print(total_by_gender, end="\n\n")


selected = customers[["Customer_ID", "Name", "Purchase_Amount"]]
print("Problem 5: Selected columns")
print(selected, end="\n\n")
```

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```
Problem 5: Customers with Age > 30
   Customer_ID     Name   Age  Gender  Purchase_Amount
1          102      Bob  35.0    Male       480.500000
2          103  Charlie  42.0    Male       320.000000
3          104    Diana  33.0  Female       150.750000
4          105    Ethan  31.0    Male       331.332222
6          107   George  47.0    Male       275.250000
7          108   Hannah  38.0  Female       530.000000
9          110    Julia  33.0  Female       365.500000

Problem 5: Total Purchase_Amount by Gender
Gender
Female    1905.240000
Male      1407.082222
Name: Purchase_Amount, dtype: float64

Problem 5: Selected columns
   Customer_ID     Name  Purchase_Amount
0          101    Alice       250.000000
1          102      Bob       480.500000
2          103  Charlie       320.000000
3          104    Diana       150.750000
4          105    Ethan       331.332222
5          106    Fiona       410.000000
6          107   George       275.250000
7          108   Hannah       530.000000
8          109      Ian       199.990000
9          110    Julia       365.500000
```

Problem 6: Data Transformation & Aggregation: Add a new column "Discounted_Price", where each customer gets a 10% discount on their Purchase_Amount. Group the data by Gender and calculate the average Purchase_Amount. Sort the dataset by Purchase_Amount in descending order.

```
customers["Discounted_Price"] = customers["Purchase_Amount"] * 0.90
print("Problem 6: With Discounted_Price")
print(customers, end="\n\n")

avg_by_gender = customers.groupby("Gender")["Purchase_Amount"].mean()
print("Problem 6: Average Purchase_Amount by Gender")
print(avg_by_gender, end="\n\n")

sorted_df = customers.sort_values(by="Purchase_Amount", ascending=False)
print("Problem 6: Sorted by Purchase_Amount (descending)")
print(sorted_df)
```

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```
Problem 6: With Discounted_Price
   Customer_ID     Name   Age  Gender  Purchase_Amount  Discounted_Price
0          101    Alice  28.0  Female       250.000000           225.000
1          102      Bob  35.0    Male       480.500000           432.450
2          103  Charlie  42.0    Male       320.000000           288.000
3          104    Diana  33.0  Female       150.750000           135.675
4          105    Ethan  31.0    Male       331.332222           298.199
5          106    Fiona  25.0  Female       410.000000           369.000
6          107   George  47.0    Male       275.250000           247.725
7          108   Hannah  38.0  Female       530.000000           477.000
8          109      Ian  29.0  Female       199.990000           179.991
9          110    Julia  33.0  Female       365.500000           328.950

Problem 6: Average Purchase_Amount by Gender
Gender
Female    317.706667
Male      351.770556
Name: Purchase_Amount, dtype: float64

Problem 6: Sorted by Purchase_Amount (descending)
   Customer_ID     Name   Age  Gender  Purchase_Amount  Discounted_Price
7          108   Hannah  38.0  Female       530.000000           477.000
1          102      Bob  35.0    Male       480.500000           432.450
5          106    Fiona  25.0  Female       410.000000           369.000
9          110    Julia  33.0  Female       365.500000           328.950
4          105    Ethan  31.0    Male       331.332222           298.199
2          103  Charlie  42.0    Male       320.000000           288.000
6          107   George  47.0    Male       275.250000           247.725
0          101    Alice  28.0  Female       250.000000           225.000
8          109      Ian  29.0  Female       199.990000           179.991
3          104    Diana  33.0  Female       150.750000           135.675
```